

# Software Requirements Specification (SRS)

## Change History Data Management Module

### Document Control

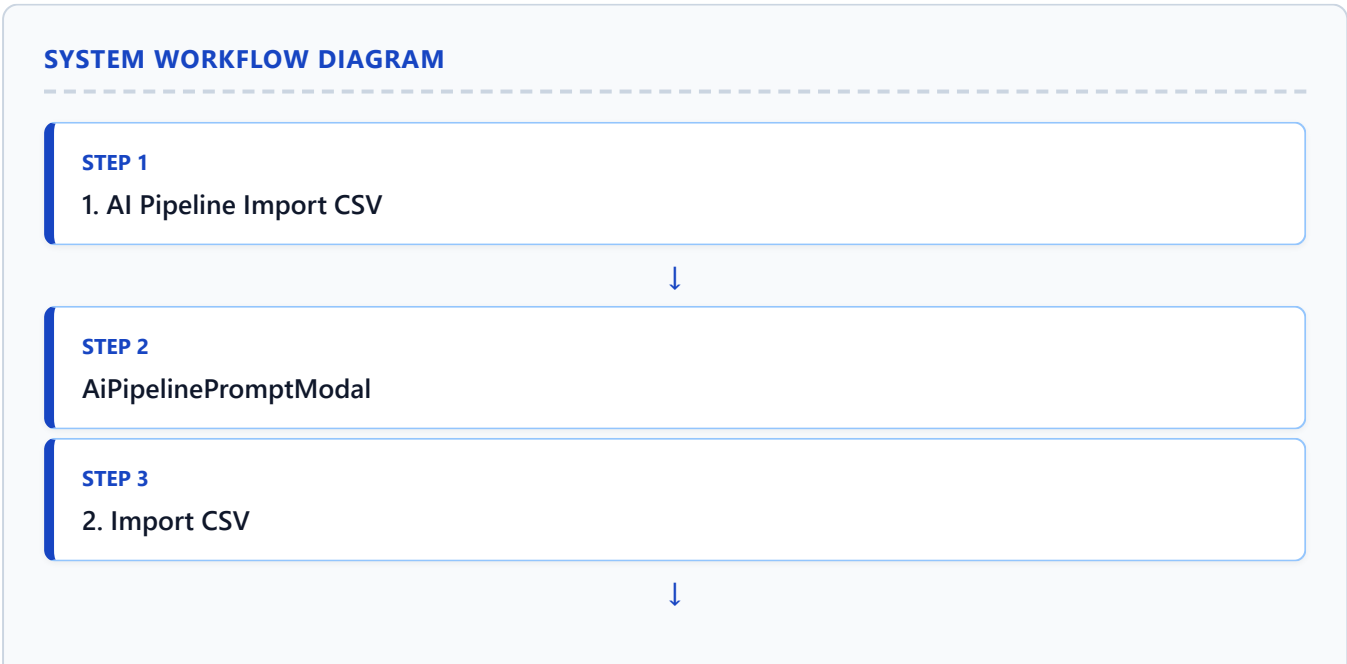
| Item             | Details                                                                    |
|------------------|----------------------------------------------------------------------------|
| Document Title   | SRS: Change History Data Management Workspace                              |
| System Module    | EQUAL Data Management Subsystem ( /data-management/change-history-data )   |
| Document Version | 1.0.0 (Pre-Implementation Baseline)                                        |
| Target Audience  | Software Engineers, QA / QC Test Engineers, Enterprise Solution Architects |

## 1. Executive Summary & Individual Flow Diagram

### 1.1 Purpose

The **Change History Data Management Module** serves as the central operational hub for managing, viewing, filtering, inline-editing, batch-deleting, importing, and exporting standardized equipment maintenance records ( `ChangeData` ).

### 1.2 Individual Module Flow Diagram



**STEP 4**

UploadDataPreview Modal

**STEP 5**

3. Export Dropdown

**STEP 6**

CSV / Excel / Targeted ZIP Export

**STEP 7**

Master Filters: Site, Process, Maint Part

**STEP 8**

Virtualized Data Grid - 22 Master Schema Attributes

**STEP 9**

Double-Click Cell Edit / Checkbox Selection

**STEP 10**

Save Button Action

**STEP 11**

Auto Trigger refreshChangeHistoryData Listener

## 2. Functional Requirements

### 2.1 Header Action Bar Sequence

- **Req-1.1 Action Button Ordering:** The top action bar MUST render buttons in strict left-to-right order:
- **AI Pipeline Import CSV**
- **Import CSV**
- **Export** (Dropdown for CSV, Excel, and Targeted ZIP)

- **Req-1.2 User Identity Tracking:** AI Pipeline import requests MUST capture and attach the authenticated operator's user name.

## 2.2 Predefined Master Data Schema (22 Attributes)

The module MUST manage records according to the 22 standardized master columns:

Site , Process , Maintenance Part , Equipment Code , Equipment Name , W/O Code , Report Content , BOM , Spare Part , Work Date , Improvement Work , Work Purpose , Problem Symptom , Problem Cause , HW Before , HW After , SW Before , SW After , Representative Work , Priority , Category , Wotype .

## 2.3 Export Functions & Targeted ZIP Package

- **Req-3.1 CSV & Excel Export:** Exports selected table rows (or all filtered rows if none selected) as .csv or .xlsx .
- **Req-3.2 Targeted ZIP Export:** When the user selects specific rows via table checkboxes, ZIP Export MUST export only the selected record IDs. If no rows are selected, it exports the default record set.

## 2.4 Automatic Data Refresh

- **Req-4.1 Navigation & Event Triggers:** Navigating to Change History or saving changes in preview modals MUST automatically trigger a data refresh so that table records and dropdown filters reflect the latest state immediately.

## 3. QC Testing & Acceptance Criteria

- **UI Button Order Test:** Verify that **AI Pipeline Import CSV**, **Import CSV**, and **Export** appear in exact order.
- **Mandatory Field Highlighting Test:** Confirm that only mandatory fields ( Site , Process , Maintenance Part , Equipment Code , Equipment Name , Representative Work , Priority , Category ) are highlighted in red when empty.
- **Targeted ZIP Export Test:** Verify that selecting row checkboxes exports a ZIP file containing only those selected records.